THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 01

Full Syllabus (Chapters 1–14) | Class 8 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** 125	$5^3 = 125$
2	**(c)** 12	$12^3 = 1728$
3	**(a)** 3^4	Subtract the exponents
4	**(d)** 1101	Repeated division of 13 by 2
5	**(a)** 95°	Angle sum of a quadrilateral is 360°
6	**(c)** parallel and equal	Basic property
7	**(c)** 3475	Ends in 0 or 5
8	**(c)** 700	$(25 \times 4) \times 7$
9	**(c)** 4200 mL	Unit conversion
10	**(a)** 1 : 8	250 g : 2000 g
11	**(a)** ₹ 3600	$SI = \frac{PRT}{100}$
12	**(d)** 210	35/100 × 600
13	**(c)** obtuse-angled	Converse idea
14	**(d)** 13	$\sqrt{25 + 144}$
15	**(d)** 2 : 3 : 5	B is already common
16	**(c)** 6	6 faces
17	**(a)** 140	Sum 700 ÷ 5
18	**(a)** $n + 2$	(3n+6)/3
19	**(a)** 8 cm	Area ÷ length
20	**(b)** square units	e.g. cm²

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$2^{10} = 1024 > 100 = 10^2$ (2).	2
22	Counting by marking a stroke for each object, grouped in fives (2).	2
23	Digit sum 11, then 2; remainder $= 2$ (2).	2
24	Reduce to simplest form (1); cross-multiply (1).	1+1
25	Other leg 15 cm (1); area 60 cm² (1).	1+1
26	126° (2).	2
27	The median also increases by 10 (2).	2
28	180 square units (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	$1372 = 2^2 \times 7^3$ (1); not a cube since 2^2 is not a triple (1); multiply by 2 (1).	1+1+1
30	$\frac{1}{2} \times 14^2 = 98$ cm² (3).	3
31	$x^2 + 11x + 18$ (2); at $x = 1$: $(3)(10) = 30 = 1 + 11 + 18$ (1).	2+1
32	SI $=$ ₹1350 (2); amount $=$ ₹8850 (1).	2+1
33	Surface area $= 6 \times 6^2 = 216$ cm² (2); 2 cubes (1).	2+1
34	$(10a + b) - (10b + a) = 9(a - b)$ (3).	3
35	Triplet $(9, 40, 41)$ (1.5); $81 + 1600 = 1681 = 1681$ (1.5).	1½+1½
	OR: Other side 9 cm (1); diagonal $\sqrt{145} \approx 12.04$ cm (2).	1+2

36	Area $= \frac{1}{2} \times 18 \times 12 = 108$ m² (2); cost ₹ 1296 (1).	2+1
	OR: Outer $24 \times 20 = 480$ (1.5); path $480 - 320 = 160$ m² (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Compare the powers of 10 first, then the decimal parts (3). (ii) $4.5 \times 10^7 = 45 \times 10^6 > 9.1 \times 10^6$ (2).	3+2
	OR: (i) $2^5 = 32$ (2). (ii) $(-2)^{3-5+1} = (-2)^{-1} = -\frac{1}{2}$ (3).	2+3
38	(i) Percentage of the value of goods or services added to it (2). (ii) ₹ 1890 (3).	2+3
	OR: (i) ₹ 135 (1). (ii) 15% (2). (iii) ₹ 688.50 (2).	1+2+2
39	(i) Mean 38.75; median $(38 + 45)/2 = 41.5$ (2). (ii) 90 (1). (iii) Large variation (2).	2+1+2
	OR: (i) Axes, plotted points, trend (3). (ii) Temperature, stock prices (2).	3+2
40	(i) 600 m² (2). (ii) Same area (1). (iii) ₹ 4800 (2).	2+1+2
	OR: (i) 10×10=100, 12×8=96, 15×5=75, 18×2=36 (3). (ii) 10 × 10 square (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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